# Physics and Mathematics of "Speka"

A Physical and Electromechanical Simulation of 1-Bit PWM PC Speaker Audio and Chassis Acoustics

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**Abstract**

This paper details the complete mathematical formulation and digital signal processing (DSP) architecture for **Speka**, an audio engine designed to physically simulate the playback of digitized audio through the internal 1-bit PC speaker of vintage personal computers (such as the IBM 5150, PC/XT, AT, and 1990s clones). Rather than treating the PC speaker as a naive bitcrusher or 1-bit square wave clipper, Speka couples four interacting physical layers: (1) the timer hardware and software modulation domain (Intel 8253/8254 PIT, RealSound PWM, and 2nd-order Sigma-Delta PDM with PolyBLEP anti-aliasing); (2) the single-ended unipolar electrical drive circuit (7406 open-collector inverter, 33 Ω ballast resistor, and inductive voice coil RL dynamics with motional back-EMF and supply rail droop); (3) the non-linear electromechanical transducer dynamics (a 2.25-inch stamped paper cone with progressive suspension hardening $K(x)$, magnetic flux decay $Bl(x)$, physical spider bottoming, and high-frequency cone breakup modes, with an alternate 12 mm ceramic piezoelectric disk model); and (4) the acoustic enclosure and chassis acoustics (internal steel case cavity resonances, front plastic slotted bezel filtering, and excursion-triggered chassis chatter and rattle).

## 1 Introduction and Historical Context

In early IBM PC-compatible architectures (commencing with the IBM 5150 in 1981), sound output was restricted to an internal miniature loudspeaker intended strictly for diagnostic beep codes. The hardware lacked a digital-to-analog converter (DAC) or programmable sound generator (PSG) such as the Texas Instruments SN76489 or General Instrument AY-3-8910 found in gaming consoles and contemporary microcomputers.

Despite these binary limitations, ingenious programmers in the late 1980s and 1990s—notably Access Software with their *RealSound* system in *Mean Streets* (1989)—devised pulse-width modulation (PWM) and pulse-density modulation (PDM) techniques that exploited the electromechanical inertia of the speaker cone as an analog reconstruction low-pass filter, enabling multi-bit digitized speech, sound effects, and music playback.

## 2 Timer Hardware and 1-Bit Modulation

### 2.1 The Intel 8253/8254 Programmable Interval Timer (PIT)

The PC speaker output is driven by Channel 2 of the Intel 8253/8254 PIT. The timer's clock input is tied to the master system oscillator divided by 12:

$$f_{\text{pit}} = \frac{14.3181818\overline{181}\text{ MHz}}{12} = 1,193,181.\overline{81}\text{ Hz} \approx 1.193182\text{ MHz} \tag{1}$$

Digital software control interfaces through system port `0x61` (PPI Port B):

- **Bit 0 (TMR2 GATE)**: Controls the count gate of Timer 2.
- **Bit 1 (SPKR DATA)**: Logically ANDed with the OUT2 output of Timer 2 to gate the signal to the 7406 driver.

### 2.2 Mode A: RealSound Pulse-Width Modulation (PWM)

In RealSound-style reproduction, Timer 2 is operated in Mode 2 (rate generator) or Mode 0 (interrupt on terminal count). The CPU reloads the counter value at an audio frame/carrier frequency $f_c \in [8\text{ kHz}, 24\text{ kHz}]$.

The number of master clock cycles per PWM carrier period is:

$$N_{\text{period}} = \left\lfloor \frac{f_{\text{pit}}}{f_c} \right\rfloor \tag{2}$$

The effective quantization bit-depth $B_{\text{eff}}$ achievable at carrier rate $f_c$ is strictly bounded by:

$$B_{\text{eff}} = \log_2(N_{\text{period}}) = \log_2\left(\left\lfloor \frac{f_{\text{pit}}}{f_c} \right\rfloor\right) \tag{3}$$

Table 1: RealSound Carrier Frequencies vs. Effective Resolution

| Carrier Frequency $f_c$ | Timer Period $N_{\text{period}}$ | Effective Bit Depth $B_{\text{eff}}$ |
|---|---|---|
| $8,000$ Hz | 149 ticks | 7.22 bits (150 levels) |
| $11,025$ Hz | 108 ticks | 6.75 bits (109 levels) |
| $16,500$ Hz | 72 ticks | 6.17 bits (73 levels) |
| $18,900$ Hz | 63 ticks | 5.98 bits (64 levels) |
| $24,000$ Hz | 49 ticks | 5.61 bits (50 levels) |

For an input audio sample $x[k] \in [-1.0, 1.0]$, the high-state pulse duration $t_{\text{high}}$ (in clock ticks) is:

$$t_{\text{high}}[k] = \text{round}\left(\frac{x[k] + 1.0}{2.0} \cdot N_{\text{period}}\right), \quad t_{\text{high}} \in [0, N_{\text{period}}] \tag{4}$$

The continuous-time duty cycle is:

$$D[k] = \frac{t_{\text{high}}[k]}{N_{\text{period}}} \tag{5}$$

### 2.3 Mode B: Pulse-Density Modulation (2nd-Order Sigma-Delta)

For demoscene tracker routines and utilities like `pcspkr`, audio is quantized into a 1-bit stream using a second-order noise-shaping Sigma-Delta ($\Sigma\Delta$) structure:

$$e[n] = x[n] - y[n-1] \tag{6}$$
$$v_1[n] = v_1[n-1] + c_1 \cdot e[n] \tag{7}$$
$$v_2[n] = v_2[n-1] + v_1[n] - c_2 \cdot y[n-1] \tag{8}$$
$$y[n] = \begin{cases} 1 & \text{if } v_2[n] \geq 0 \\ 0 & \text{if } v_2[n] < 0 \end{cases} \tag{9}$$

where $c_1 = 0.8$ and $c_2 = 0.5$ ensure stability while pushing quantization noise into the supersonic region.

### 2.4 Anti-Aliased Edge Synthesis via PolyBLEP

Direct sampling of discontinuous 1-bit pulses at digital audio rates ($f_s = 44.1$ kHz–96 kHz) causes severe foldover aliasing. To avoid this, transitions between binary levels $y \in \{0, 1\}$ are smoothed using a second-order Polynomial Band-Limited Step (PolyBLEP) residual $R(t)$:

$$R(t) = \begin{cases} 2t - t^2 - 1 & \text{for } 0 \le t < 1 \\ t^2 + 2t + 1 & \text{for } -1 \le t < 0 \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

where $t = \frac{\tau - \tau_{\text{edge}}}{T_s}$ is the normalized time offset from the sampling instant. For every rising step of magnitude $\Delta y = +1$, $R(t)$ is added; for every falling step $\Delta y = -1$, $R(t)$ is subtracted.

### 2.5 CPU Interrupt Latency and DRAM Refresh Jitter

On real x86 systems, DRAM refresh cycles (which steal $\sim 5\%$ of bus memory cycles via DMA Channel 0 every 15.1 $\mu$s) and interrupt servicing create jitter in timer reload times:

$$t_{\text{jitter}}[n] = \sigma_j \cdot w[n] \tag{11}$$

where $w[n] \sim \mathcal{N}(0, 1)$ is Gaussian noise and $\sigma_j \approx 2$–$5\,\mu$s. This reproduces the characteristic low-level background "grate" and flutter of PC speaker audio.

## 3 Electrical Circuit and Driver Stage

### 3.1 Unipolar 7406 Open-Collector Drive

The speaker in the IBM 5150 is connected between the +5V DC supply rail and the collector output of an open-collector inverter (7406) via a 33 $\Omega$, 0.5W series current-limiting resistor $R_{\text{ballast}}$:

$$R_{\text{ballast}} = 33\,\Omega, \quad R_e \approx 8\,\Omega, \quad L_e \approx 0.2 \text{ mH} \tag{12}$$

When the 7406 is ON (output pulled to ground):

$$I_{\text{peak}} \approx \frac{V_{cc} - V_{\text{sat}}}{R_{\text{ballast}} + R_e + R_{\text{on}}} = \frac{5.0 - 0.2}{33 + 8 + 0.5} \approx 115 \text{ mA} \tag{13}$$

When the 7406 is OFF: the collector is high-impedance ($I \to 0$).

**Critical Insight (Unipolar Excursion Bias):** The voice coil is driven **strictly unipolar**. Current never reverses. Consequently, at a 50% duty cycle, the voice coil experiences a persistent DC bias current of $\bar{I} \approx 57$ mA, shifting the cone equilibrium position forward:

$$x_{\text{DC}} = \frac{Bl_0 \cdot \bar{I}}{k_0} \approx \frac{1.2 \cdot 0.057}{3400} \approx 0.02 \text{ mm} \tag{14}$$

### 3.2 Voice Coil RL State-Space with Motional Back-EMF

The electrical current $i(t)$ through the voice coil satisfies:

$$\frac{di}{dt} = \frac{1}{L_e} \left[ V_{\text{drive}}(t) - (R_{\text{ballast}} + R_e + R_{\text{sw}}(t))\, i(t) - e_b(t) \right] \tag{15}$$

where:

- $R_{\text{sw}}(t) = 0.5\,\Omega$ when the 7406 gate is active ($y = 1$), and $R_{\text{sw}}(t) = R_{\text{off}} \approx 10^5\,\Omega$ when off ($y = 0$).
- $e_b(t) = Bl(x) \cdot v(t)$ is the motional back-EMF generated by cone velocity $v(t) = \dot{x}(t)$.
- $V_{\text{drive}}(t) = V_{cc} - \Delta V_{\text{droop}}(t)$ models power supply rail sag under sustained current demand.

### 3.3 Power Supply Rail Sag

The +5V rail droops slightly when current is drawn through the speaker circuit:

$$\frac{d\Delta V_{\text{droop}}}{dt} = \frac{1}{\tau_{\text{droop}}} \left(\gamma_{\text{sag}} \cdot i(t) - \Delta V_{\text{droop}}(t)\right) \tag{16}$$

with recovery time constant $\tau_{\text{droop}} \approx 15$ ms and sag coefficient $\gamma_{\text{sag}} \approx 2.5$ V/A. This creates dynamic compression on dense pulse trains.

## 4 Electromechanical Speaker Cone Dynamics

### 4.1 Non-Linear Thiele-Small Equations of Motion

The mechanical motion of the 2.25-inch paper cone is modeled as a coupled second-order non-linear mass-spring-damper system:

$$\frac{dx}{dt} = v(t) \tag{17}$$

$$\frac{dv}{dt} = \frac{1}{M_{ms}} \left[F_{\text{Lorentz}}(t) - R_{ms} v(t) - K(x) \cdot x(t)\right] \tag{18}$$

where:

- $M_{ms} = 0.7 \times 10^{-3}$ kg (moving mass of cone, voice coil, and co-moving air).
- $R_{ms} = 0.35$ N $\cdot$ s/m (mechanical suspension damping, yielding $Q_{ms} \approx 4.4$).
- $F_{\text{Lorentz}}(t) = Bl(x) \cdot i(t)$ is the electromagnetic driving force.

### 4.2 Nonlinear Suspension Hardening $K(x)$

As cone excursion approaches the spider and surround limits ($X_{\max} \approx \pm 0.8$ mm), the restoring spring force stiffens dramatically:

$$K(x) = k_0 \cdot \left(1 + \beta_2 x^2 + \beta_4 x^4\right) \tag{19}$$

where $k_0 = 3400$ N/m sets the small-signal fundamental resonance:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_0}{M_{ms}}} \approx \frac{1}{2\pi}\sqrt{\frac{3400}{0.0007}} \approx 350.8 \text{ Hz} \tag{20}$$

and $\beta_2 = 4 \times 10^6$ m$^{-2}$, $\beta_4 = 1.6 \times 10^{13}$ m$^{-4}$. When excursion exceeds $X_{\max}$, an inelastic mechanical boundary condition is applied:

$$v(t^+) = -\eta \cdot v(t^-), \quad \eta = 0.12 \tag{21}$$

producing hard mechanical "bottoming" clicks.

### 4.3 Magnetic Force Factor Decay $Bl(x)$

When the voice coil leaves the uniform magnetic field of the permanent magnet gap:

$$Bl(x) = Bl_0 \cdot \exp\left(-\frac{x^2}{2\sigma_{\text{gap}}^2}\right) \tag{22}$$

with $Bl_0 = 1.2$ N/A and $\sigma_{\text{gap}} = 0.6$ mm.

### 4.4 Far-Field Radiated Sound Pressure

For a miniature unbaffled or small-baffled driver, sound pressure in the acoustic far field is proportional to **cone acceleration**, not displacement:

$$p_{\text{acoustic}}(t) = \frac{\rho_0 S_d}{2\pi r} \cdot \frac{d^2x}{dt^2} = \frac{\rho_0 S_d}{2\pi r} \cdot \frac{dv}{dt} \tag{23}$$

where $\rho_0 \approx 1.204$ kg/m$^3$ is air density and $S_d \approx 2.5 \times 10^{-3}$ m$^2$ is effective cone area.

### 4.5 Cone Breakup Modes

At frequencies above 1.5 kHz, the paper diaphragm ceases to move as a rigid piston, exhibiting flexural bending modes. We model three modal resonators in parallel:

$$H_{\text{breakup}}(s) = \sum_{m=1}^{3} \frac{g_m \omega_m^2}{s^2 + \frac{\omega_m}{Q_m} s + \omega_m^2} \tag{24}$$

with:

- Mode 1: $f_1 = 2,400$ Hz, $Q_1 = 3.2$, $g_1 = 0.35$
- Mode 2: $f_2 = 3,750$ Hz, $Q_2 = 4.0$, $g_2 = 0.25$
- Mode 3: $f_3 = 5,100$ Hz, $Q_3 = 2.8$, $g_3 = 0.15$

### 4.6 Alternative Transducer: Ceramic Piezoelectric Disk

For budget clone motherboards and laptops (e.g. ThinkPad 700 series), the driver is a 12 mm ceramic piezo buzzer:

$$H_{\text{piezo}}(s) = \left(\frac{s}{s + \omega_{\text{hp}}}\right)^2 \cdot \frac{\omega_p^2}{s^2 + \frac{\omega_p}{Q_p} s + \omega_p^2} \tag{25}$$

with high-pass cutoff $f_{\text{hp}} = 1,200$ Hz (12 dB/octave low-end cutoff) and prominent ceramic resonance $f_p = 3,250$ Hz ($Q_p = 8.5$).

## 5 Enclosure and Chassis Acoustics

### 5.1 Internal Metal Case Cavity Modes

The speaker's rear wave radiates into the steel enclosure (dimensions $\sim 45 \times 40 \times 15$ cm). The internal standing waves and metallic reflections are represented as an un-damped resonator bank:

$$H_{\text{chassis}}(s) = 1 + \sum_{k=1}^{4} \frac{a_k \omega_k s}{s^2 + \frac{\omega_k}{Q_k} s + \omega_k^2} \tag{26}$$

with frequencies $f_k \in \{185 \text{ Hz}, 340 \text{ Hz}, 520 \text{ Hz}, 790 \text{ Hz}\}$ and $Q_k \in [3.5, 6.0]$.

### 5.2 Front Plastic Bezel Grille Damping

The sound radiating forward traverses narrow horizontal ventilation slots in the ABS front fascia:

$$H_{\text{bezel}}(s) = \frac{\omega_b^2}{s^2 + \frac{\omega_b}{Q_b}s + \omega_b^2}, \quad f_b = 4,500 \text{ Hz},\ Q_b = 0.707 \tag{27}$$

### 5.3 Excursion-Triggered Chassis Rattle and Screw Chatter

Vibration from large cone displacements excites chassis panel chatter:

$$E_{\text{rattle}}[n] = (1 - \alpha_r)E_{\text{rattle}}[n-1] + \alpha_r \cdot \max(0, |x[n]| - X_{\text{rattle_thresh}}) \tag{28}$$

When $E_{\text{rattle}}[n] > 0$, an impulsive stochastic noise burst is excited:

$$rattle[n] = E_{\text{rattle}}[n] \cdot \xi[n] \cdot \text{sign}(v[n]) \tag{29}$$

where $\xi[n]$ is uniform white noise bandpass-filtered between 800 Hz and 3.5 kHz.

## 6 Numerical Discretization and Real-Time Integration

The coupled state system $[\Delta V_{\text{droop}}, i, x, v]^T$ is updated using a second-order Runge-Kutta (Heun's method) or semi-implicit Euler integration scheme at the sampling period $T_s = 1/f_s$. Because typical audio sample rates (44.1 kHz to 96 kHz) provide time steps $T_s \approx 10$–$22\,\mu$s, a $2\times$ or $4\times$ internal sub-stepping loop is used for the stiff electrical voice coil equation ($L_e = 0.2$ mH, $R \approx 41\,\Omega \implies \tau_e = 4.8\,\mu$s):

$$\Delta t = \frac{T_s}{M_{\text{substeps}}}, \quad M_{\text{substeps}} = 4 \tag{30}$$

This guarantees numerical stability without requiring implicit matrix solvers.

## 7 Conclusion

The resulting **Speka** system provides an authentic physical model capturing:

1. The characteristic high-pitched carrier whine and timer-step distortion of vintage RealSound PWM routines.
2. The unipolar asymmetric forward-cone excursion bias and voice coil motional back-EMF.
3. The thin, boxy 350 Hz cone resonance, high-frequency paper breakup modes, and complete absence of sub-bass.
4. The metallic chassis ringing and mechanical chatter characteristic of vintage beige desktop personal computers.